DILPREET KAUR

AI/ML Engineer

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SUMMARY

B.Tech Information Technology student (2027) building AI/ML solutions. Former Machine Learning Intern at NIC with experience in end-to-end AI applications.

EDUCATION

Guru Tegh Bahadur Institute of Technology, Delhi 2023 – 2027
Bachelor of Technology in Information Technology, CGPA: 8.38

SS Mota Singh School, Delhi 2011 – 2023
Class X: 86%, Class XII: 80%

EXPERIENCE

Machine Learning Intern, National Informatics Centre (NIC) Jun 2025 – Aug 2025

- Cleaned multi-gigabyte log datasets with Pandas and regex; built a rule-based labeling system for 7+ log categories (port scans, DDoS, failed SSH, malware) to bootstrap training labels.
- Fine-tuned a T5-small transformer for automated security-log summarization on 500 labeled logs, reaching 0.91 ROUGE-1/ROUGE-L on validation.
- Built BERT-based classification pipelines to sort logs into 3 categories (Error, Warning, Information), streamlining triage for downstream analysis.
- Diagnosed a 0.91 to 0.45 ROUGE drop between validation and full-dataset inference and proposed GPT-4-assisted relabeling to improve generalization.

PROJECTS

IT Support Copilot, Agentic RAG System *Live Demo*

- Built an agentic RAG copilot for a simulated 3,000-employee company using LangGraph to route each query between direct chat, a private Pinecone knowledge base, and Tavily web search, with evidence grading, query rewriting, and a retry guard that returns "insufficient evidence" instead of hallucinating.
- Developed FastAPI endpoints for chat and admin-protected document ingestion (PDF, DOCX, TXT, MD) so new policies become searchable immediately; added a decision trace, SQLite audit logging, and Docker packaging; deployed on Vercel.
- **Tools:** Python, LangGraph, FastAPI, Pinecone, Groq, Tavily, SQLite, Docker, Vercel

AI Travel Planning Agent, LangGraph with Long-Term Memory *Live Demo*

- Built a multi-agent travel planner in LangGraph that generates personalized itineraries from destination, duration and budget.
- Integrated Tavily for travel research, Groq Llama 3.3 for reasoning, and PostgreSQL checkpointing for memory across sessions.
- Generated a 7-day Japan travel itinerary within a 2 lakh INR budget using coordinated agent workflows.
- **Tools:** Python, LangGraph, LangChain, PostgreSQL, Groq, Tavily

GridScan RL, Reinforcement Learning Environment *Live Demo*

- Built a tabular Q-learning agent (Bellman update, epsilon-greedy with decay) that scans a 10x10 obstacle grid using three actions, reaching 90.1% coverage (82 of 91 free tiles).
- Designed shaped rewards and cycle detection, tracking coverage, collisions and loops per episode.
- Built a FastAPI and NumPy engine with unit tests and a React dashboard showing live Q-values.
- **Tools:** Python, NumPy, FastAPI, React, Q-learning

AI-Powered Deep Packet Inspection and Threat Detection *Live Demo*

- Trained Random Forest models on 32 engineered features to classify applications and detect 5 threat classes.
- Developed a multithreaded C++ PCAP analyzer that identifies applications via TLS SNI and HTTP Host.
- Built a Streamlit dashboard presenting traffic insights and threat scores.
- **Tools:** C++, Python, Scikit-learn, Pandas, Streamlit

SKILLS

- **Machine Learning:** Supervised Learning, Classification, Random Forest, LightGBM, Scikit-learn, Reinforcement Learning (Q-learning), Feature Engineering, Hyperparameter Tuning, Model Evaluation, MLflow
- **Deep Learning and NLP:** Neural Networks, Transformers (T5, BERT), Hugging Face, Fine-tuning, Text Summarization, Text Classification, Embeddings
- **LLM and Agents:** LLM Applications, Prompt Engineering, Multi-Agent Systems (LangGraph, LangChain)
- **Programming and Data:** Python, C++, SQL, PostgreSQL, SQLite, Pandas, NumPy
- **Backend and Deployment:** FastAPI, REST APIs, Streamlit, Gradio, Git, GitHub

ACHIEVEMENTS AND CERTIFICATIONS

- Secured 1st Position at Perfect Pitch 2.0 Ideathon (E-Cell GTBIT x NASSCOM Foundation).
- Open-source contributor, GirlScript Summer of Code (GSSoC).
- **Certifications:** Microsoft Generative AI, IBM Machine Learning with Python.